Fiscal Year:	FY 2024	**Task Last Updated:**	FY 12/26/2024
PI Name:	Cao, Xu Ph.D.		
Project Title:	Identifying Genetic Factors in Radiation Injury with Pooled Single Cell Sequencing		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	TRISH--TRISH		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	94305	**Congressional District:**	18
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2022 TRISH-RFA-2201-PD: Translational Research Institute for Space Health (TRISH) Postdoctoral Fellowships
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No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	TRISH
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Wu, Joseph M.D., Ph.D. (MENTOR: Stanford University)		
Grant/Contract No.:	NNX16AO69A-P0701		
Performance Goal No.:			
Performance Goal Text:			
	POSTDOCTORAL FELLOWSHIP An astronaut's heart and vessels can be damaged by high energy space radiation during prolonged space travel out of the protection of Earth's magnetic fields. These damages are consequences of radiation-induced heart inflammation, and therefore fibrosis (similar to scarring), which impedes heart contraction. The major aim of the proposed study is to find out how and which inheritable/genetic factors may affect people's susceptibility to space radiation. This information can help us better predict the risk of heart diseases in astronauts and design personalized drugs targeting the identified genetic factor. To identify the underlying genetic factor, we need to perform an association study between genetic factors and human		

Task Description:	heart damages within a large population exposed to radiation. However, radiated human heart samples are extremely difficult to obtain and substantially affected by undesired factors (age, lifestyle, health status), making them unavailable and not ideal for this study. Human-induced pluripotent stem cells (iPSCs), derived from patients, can give rise to heart-like tissues in culture dishes, which can be an alternative to human heart samples. These iPSC-derived heart tissues can be batch produced and maintained in standardized condition, which could eliminate undesirable factors and keep the information of individuals' genetic backgrounds. In the first part of this study, we will obtain 100 different iPSC lines derived from a diverse population with equal distribution of ethnicities and genders. Fortunately, these lines have already been generated by the Stanford Biobank, led by Dr. Joseph Wu (mentor). All lines will be pooled together to be induced toward immune cells (macrophages) and then irradiated. Next, using single cell sequencing technology, we can distinguish the line identity based on the reference genome and find out the radiation-induced damage/inflammation in each line. Last, we will do a correlation study to figure out which genetic factors could affect the radiation response of immune cells. In the second part of the study, we will validate the contribution of identified genetic factors in radiation-induced heart damages in multiple systems, including immune cells (macrophages) and 3D heart tissues generated from iPSCs. For the immune cell model, we will compare radiation-induced immune responses of macrophages with and without the candidate genetic factor. With the 3D heart tissue model, we will investigate whether the presence of the specific genetic factor could affect their beating and fibrosis (scar formation) after radiation exposure. Successful completion of this study will provide us (i) the role of inheritable genetic factors in space radiation-induced heart disorders, (ii) a new pre-screening platform for heart disease risks of astronauts, (iii) novel druggable targets for the protection of radiation-caused heart damages. Reducing uncertainties in cardiovascular risks associated with space radiation will accelerate humanity's dream to travel in space.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	The key findings and their impact from the second year of this project are summarized as follows: 1. scRNA-seq of the Cell Village system as a novel tool: Our use of single-cell RNA sequencing (scRNA-seq) on pooled iPSC lines within the Cell Village system has proven to be an innovative and cost-effective method for studying individual-specific responses to radiation. This approach allows us to investigate the genetic and cellular mechanisms underlying radiosensitivity across a diverse set of iPSC lines in a scalable manner, providing insights at a fraction of the cost of traditional methods. 2. Technical challenges in the Cell Village method: One of the significant technical hurdles we encountered is the unequal distribution of cell numbers from different individuals within the pooled system. This imbalance can potentially affect data interpretation, as some individuals are underrepresented in the scRNA-seq data. Optimizing the protocol to ensure more equitable cell representation across all lines will be critical for enhancing the robustness of our findings. 3. Validation of SNPs through eQTL analysis: The SNPs identified from the eQTL analysis of our dataset using 20 iPSC lines align well with those previously reported in much larger datasets. This consistency underscores the accuracy of our scRNA-seq approach and the validity of the Cell Village system for detecting genetic variations linked to radiation responses. It also demonstrates that the smaller scale of our study can still yield meaningful results when compared to broader datasets. 4. Complementary role of the Cell Village approach to clinical GWAS: The Cell Village system serves as a valuable complement to clinical genome-wide association studies (GWAS). By validating previously reported single nucleotide variants (SNVs) associated with radiosensitivity, this approach bridges the gap between in vitro cellular models and clinical findings, allowing for a more comprehensive understanding of the genetic factors influencing radiation responses. 5. Potential for novel SNV discovery: With an increase in the number of iPSC lines included in future studies, the Cell Village approach holds significant potential for identifying novel SNVs associated with radiosensitivity. Expanding the diversity of the cell lines will enhance the statistical power of the analysis, facilitating the discovery of new genetic variants that may play a crucial role in individual-specific radiation responses. These findings highlight the Cell Village system's promise as both a research tool for studying radiosensitivity and a cost-effective method to complement and extend the results of clinical studies.
Bibliography Type:	Description: (Last Updated: 03/12/2025)
Articles in Peer-reviewed Journals	Whitcomb LA, Cao X, Thomas D, Wiese C, Pessin AS, Zhang R, Wu JC, Weil MM, Chicco AJ. "Mitochondrial reactive oxygen species impact human fibroblast responses to protracted γ-ray exposures." International Journal of Radiation Biology. 2024 Jun 2;100(6):890-902. https://doi.org/10.1080/09553002.2024.2338518 ; PMID: 38631047; PMCID: PMC11471570 , Jun-2024
Articles in Peer-reviewed Journals	Cao X, Thomas D, Whitcomb L, Wang M, Chatterjee A, Chicco AJ, Weil MM, Wu JC. "Modeling ionizing radiation-induced cardiovascular dysfunction with human iPSC-derived engineered heart tissues." Journal of molecular and cellular cardiology. 2024 Mar;188:105. https://doi.org/10.1016/j.yjmcc.2023.11.012 ; PMID: 38431383; PMCID: PMC10961094 , Mar-2024
Awards	Cao X. "Stanford Cardiovascular Institute Travel Award, June 2024." Jun-2024